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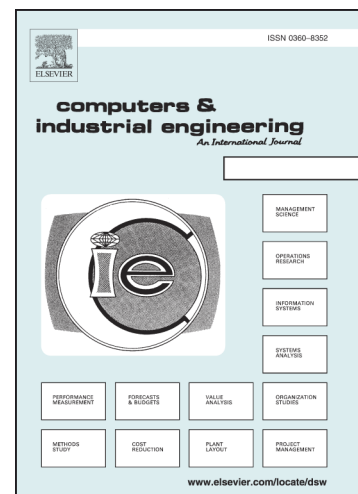
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**A bi-objective model for location planning of electric vehicle charging
stations with GPS trajectory data**

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Abstract

The construction of charging stations is a crucial factor in promoting electric vehicles (EV). It is necessary to construct EV charging stations in advance to encourage drivers to prefer EVs. This paper addresses the EV charging stations location problem in a city with low EV penetration rate. We divide the city into a grid with several same cells. The potential charging demand of each cell is estimated with the use of GPS trajectory data from thousands of traveling vehicles in the network. We present a cell-based model to decide locations, capacity options, and service types for EV charging stations that can cover all potential charging demand. The problem is formulated as a bi-objective mixed-integer mathematical model, with one objective related to minimizing cost and the other related to maximizing service quality. To solve it, we propose a hybrid evolutionary algorithm that combines the non-dominated sorting genetic algorithm-II (NSGA-II) with linear programming and neighborhood search. We conduct computational experiments on randomly generated instances to evaluate the performance of the proposed hybrid NSGA-II. Finally, we present a case study designing an EV charging station network for Shenzhen, China with real GPS trajectory data. We also offer some management insights of EV charging stations construction based on sensitivity analysis.

Keywords: Electric vehicles; Charging infrastructure location; Bi-objective optimization; Bi-objective Evolutionary algorithm; GPS trajectory data

1. Introduction

The increasing growth of petroleum-based vehicles leads to huge amounts of fossil fuel consumption and greenhouse gas emissions around the world. Due to resource and environmental concerns, governments and people are looking to alternative fuel vehicles. EVs have attracted a great deal of enthusiasm as a good replacement for petroleum-based vehicles, which accounts for 81% of the total alternative fuel vehicles in China. Nevertheless, the penetration level of EVs is only 0.5% of the total motor vehicles in China. The lack of charging infrastructure in the road networks is one of the major barriers to promote EVs (Chung and Kwon, 2015). Thus, the government encourages constructing charging infrastructures in advance to cover potential charging demand and attract more EV users. The aim of this paper is to study how to locate EV charging stations in a city with low EV penetration.

The electric charging station location problem is a kind of refueling station location problem. The model of demand plays a key role in refueling station location problem. There are two major approaches to deal with demand. The first is to aggregate demand in fixed

locations, such as nodes or links (Nicholas et al. 2004). The population is always used as a relative estimator of demand. This way may not be suitable for refueling station location problem because drivers always produce demand when traveling. The second type uses passing flows to express demand (Hodgson, 1990; You and Hsieh, 2014). It may be more realistic to model the demand as flows in the network. However, Origin-Destination (O-D) flow data is difficult to collect from the vehicles traveling in the network (Lin et al. 2008), especially for sparse EVs. Further, the sale of EVs is predicted to grow rapidly in the following years. It is necessary to consider potential charging demand in the location of charging stations. Therefore, we propose a cell-based location model for electric charging station location problem. The city transportation network is divided into a grid with several same cells. A cell is a minimum area that uses to estimate potential charging demand and locate stations.

The cell-based approach shows the following advantages. First, we propose a method to estimate the potential EV charging demand based on the current traffic state. We assume that the area with more petroleum-based vehicles at present will have more EVs in the future, and then produce more charging demand. GPS trajectory data, which can provide a good understanding of the current traffic situation, is used to estimate the potential charging demand. Second, the cell-based assumption is more conformed to city planning rules. In the city facility planning, the decision makers need to take account of the locations of various facilities at the same time. Previous models, such as flow-based models and node-based models, try to find the most appropriate points to locate facilities in the network. They do not consider whether these points are vacant or not. The cell-based solution provides the best cells to locate stations, which offers the decision makers more flexibility. Third, the cell-based model is based on the set covering problem. At the initial stage of constructing charging stations, the decision makers may want to know the minimum number of charging stations that can serve all potential charging demand. The proposed cell-based model can answer this question. Furthermore, two objective functions are included: minimize total cost and maximize service quality. A set of location solutions with different cost and service quality is provided to decision makers for reference.

The multiple types of charging services and locating modes are the distinctive features of the location of electric charging stations. From the perspective of charging architecture or energy supply mode, the EV charge can be classified as plug-in charging, battery swapping (Liu, Wang and Wu et al., 2016) and wireless charging. For plug-in charging, there are three levels of charging technologies from the perspective of power level: slow (level 1), medium (level 2), and fast (level 3) charging (Morrow, Karner, and Francfort, 2008; Bandyopadhyay et al., 2012). For levels 1 and 2 charging technologies, drivers need to wait 2-8 hours to fully charge their EVs, which could only be located in residential parking areas. Level 3, fast plug-in charging, can charge vehicles to 80% capacity within 30 minutes. Battery swapping only takes 5-10 minutes. Furthermore, there are three locating modes of charging stations: parking-lot-based, gas-station-based and random-based (Cai et al., 2014). Parking-lot-based charging stations are mostly located at the parking spaces in residential communities or

shopping malls. They usually provide normal charging service because drivers are likely to stay at these places for a long time. Gas-station-based and random-based charging stations mainly support fast plug-in charging or battery swapping services to EVs, which offer higher convenience and accessibility. In the design of the charging station network, decisions on service types and locating modes of stations should be considered because they are strongly related to cost and service quality.

Based on the above, we focus on locating two types of charging stations: battery-swapping stations and fast plug-in charging stations in this study. In practice, the operation of a battery-swapping station is much more complex than that of a fast plug-in charging station because additional batteries are needed to support battery swapping. Some companies and researchers have proposed new business models for battery swapping, for example, battery-leasing (Mak et al., 2013). *NIO*, an EV company in China, has launched a project of constructing an EV charging station network in China that offers fast plug-in charging and battery swapping services. We assume that such new strategies would be widely accepted by consumers and investors.

In this paper, we present a cell-based model to decide locations, capacity options, and service types for EV charging stations that can cover all potential charging demand. The city area is divided into a grid of the same cells with different charging demand, which is estimated from GPS trajectory data. The problem is formulated as a bi-objective mixed integer linear programming (MILP) model with the two objectives of minimizing total cost and maximizing service quality. The proposed model is based on the set covering problem, which is formally NP-hard (ReVelle and Eiselt, 2005). The bi-objective model requires a set of Pareto solutions, which makes the problem more difficult to solve. For these reasons, we propose a hybrid evolutionary algorithm that combines the NSGA-II with linear programming (LP) and neighborhood search. The proposed method extends the NSGA-II to deal with the problem involving both discrete and continuous decision variables. Finally, we present a case study designing an EV charging station network in Shenzhen, China using GPS trajectory data from thousands of vehicles. We also offer some management insights based on numerical experiments into the construction of EV charging stations.

The remainder of this paper is as follows. In section 2, a relevant literature review is presented. Next, a mathematical model of the problem is presented in section 3. The solution method is proposed in section 4 to obtain efficient Pareto set. In section 5, we evaluate the performance of the proposed algorithm and present a case study in Shenzhen, China. The paper closes with conclusions in section 6.

2. Literature review

Facility location problem refers to sitting facilities in some given space with various objectives. The applications of location problems range from gas stations and fast food outlets to landfills and power plants (ReVelle and Eiselt, 2005). Refueling station location problem is one of the extensively studied facility location problems, among which EV charging station location problem attracts more attention due to the increasing environmental

requirements (Lam et al., 2014; Shahraki et al., 2015; Arslan and Karasan, 2016). We classify the refueling location problem in terms of demand modeling approaches.

One approach that studies refueling station location problem is based on path flow demand, which was first proposed by Hodgson (1990) called flow capturing location-allocation model (FCLM). The flow-based model was to select locations of p stations in a network to maximize captured flows. The basic input of the flow-based models is a set of O-D pairs and traffic volumes between them. It assumed that if the OD flow passed through a facility on the path, it could be captured by that facility. Many papers extended this approach in terms of modeling (MirHassani and Ebrazi, 2013; Kim and Kuby, 2012; Kuby and Lim, 2005; Kuby and Lim, 2007; Upchurch et al., 2009; Wang and Lin, 2009) and solution methods (Yıldız et al., 2015; Lim and Kuby, 2010). Based on the flow-based model, Chung and Kwon (2015) formulated a multi-period optimization model for EV charging station location and Wang and Lin (2013) extended it with consideration of multiple services types. The flow-based models express refueling demand by passing flows that seems more realistic. However, the lack of EV data and unordered flows in the inner city road network make it impossible to collect or estimate EV flows. The flow-based models are suitable for the network with ordered flows, such as expressway network or railway network.

Another approach is based on node demand, using P-median location model (Hakimi, 1964) or set covering model (Toregas et al., 1971). Nicholas et al. (2004), Nicholas and Ogden (2006) used this approach to locate refueling stations. These studies used population distribution data to estimate demand. This assumption may not suitable for refueling station location problem because drivers always produce demand when traveling. Lin et al. (2008) supposed that any point along a road with higher VMT (vehicle miles traveled) was more likely to be an origin of the refueling trip. They quantified the refueling demand of a node with the use of VMT data. The objective of the model was to minimize the travel time drivers reaching the closest refueling stations. Other methods investigated EV charging station planning by considering one or part of the above factors, such as simulation (Cai et al., 2014; Xi et al., 2013), activity-based assessment (Dong et al., 2014), robust optimization (Mak et al., 2013). To the best of our knowledge, the optimization model including decisions on locations, service types, locating modes and capacity options has not been studied in the literature.

In the above traditional models, the final purpose of most models is to decide the locations of refueling stations. EV charging station location planning includes not only the location of charging stations, but also the arrangement of service types, locating modes and capacity options. Moreover, data used in the traditional models is difficult to collect, such as OD path flow data, which limits their applications in real infrastructure planning. Finally, the balance of different objectives has not been sufficiently studied. Most traditional models are single-objective, with the objective of either minimizing total cost or maximizing cover.

Unlike other studies, our model is based on cell demand. The city area is divided into a grid with charging demand in each cell, which is estimated from GPS trajectory data. We develop a bi-objective model to balance cost and service quality for locating charging stations,

considering two different types of location (gas-station-based and random-based) and two types of charging services (fast plug-in charging and battery swapping). In the problem, not only the optimal locations of charging stations are determined, but also the optimal service types and optimal capacity options are arranged. The objective functions include: 1) minimizing the total cost, 2) minimizing the time spent for a random driver to complete charging from the point where the driver has a charging demand.

A vast variety of techniques has been proposed to find the Pareto set of multi-objective optimization problems. The most common methods are to transform multiple objectives into a single objective function, which are known as the scalarization methods, such as weighted-sum and ϵ -constraint (Ehrgott, 2006). The solution of one instance of scalarization methods is a efficient solution of the original multi-objective problem (Caramia and Dell'Olmo, 2008). To obtain efficient Pareto set, the scalarization methods require solving multiple instances of the problem and may take too much computational time. Hence, they might not be applicable tools for large problems. Meta-heuristics techniques are more efficient for larger and more complex problems because they can obtain an approximate Pareto set in one run. Farahani et al. (2010) reviewed recent studies using meta-heuristics techniques to find Pareto sets of multi-objective location problems. They concluded that the genetic algorithm class seemed to be more appropriate for multi-objective location problems. NSGA-II (Deb et al., 2002) is one of the most powerful multi-objective meta-heuristic genetic algorithms. It has been widely used to solve multi-objective optimization problems in different areas, such as public facility location (Doerner et al., 2009), hospital waste management network design (Medaglia et al., 2009), supply network design (Villegas et al., 2006) and congested facility location (Hajipour et al., 2015). In most above papers, the NSGA-II based methods outperform other methods in the solution quality. Therefore, we adopt and improve the NSGA-II to solve the proposed problem in this paper.

3. Problem formulation

This section introduces a bi-objective location model for locating EV charging stations. The associated indices and input parameters are listed in Table 1 and 2 respectively.

3.1 Problem description

The bi-objective charging station location problem is defined as follows: Given an urban transportation network, decide locations, capacity options, and service types for EV charging stations that can cover all potential charging demand. The objectives are minimizing the total cost and maximizing the service quality. The budget for infrastructure deployment is a crucial issue for decision makers who are seeking to minimize the total cost to them. To achieve more profits over the long term, maximizing service quality should also be an alternative objective. Obviously, the construction cost of random-based stations is higher than that of gas-station-based stations because of new building fees. Nevertheless, candidate sites for gas-station-based location are limited, which reduces the probability of ensuring a good coverage. In addition, fast plug-in charging has longer service time but lower cost, while battery

swapping has shorter service time but needs higher expenditure. Therefore, the two objectives, minimizing cost and maximizing service quality, are essentially in conflict.

Table 1 Indexes

i	Index of a demand cell, $i \in \{1, \dots, N\}$
j	Index of a candidate cell, $j \in \{1, \dots, N\}$
h	Index of a capacity option, $h \in \{1, \dots, H_j\}$
k	Index for the service type of a charging station, $k \in \{1, \dots, K\}$
	$k=1$ represents a fast-charging station, $k=2$ represents a battery-swapping station

Table 2 Input parameters

N	Number of cells in the grid, i.e. the number of candidate and demand sites
K	Number of service types, $K=2$
H_j	Maximum number of capacity units a station can provide in cell j
p	The capacity unit, which denotes the minimum level of the station capacity option. The capacity options of a station in candidate cell j can be $0, p, 2p, \dots, H_j p$
f_i	Number of vehicles in cell i
d_i	Charging demand in cell i
v_i	Average travel speed of cell i
ρ_i	The density of road network in cell i
μ_i	Area of cell i
c_{jk}	Cost of constructing a new charging station of type k in cell j with one capacity unit
δ	The rate of the cost increase when the capacity of the station goes up, $0 < \delta < 1$
e	Saved cost of constructing a gas-station-based charging station
b_j	Equals 1 if there is a gas station in cell j ; otherwise, equal to 0
F	Total number of the charging demand in the network, $F = \sum_{i=1}^N d_i$
t_{ij}	Travel time from a demand cell i to a candidate cell j
s_k	The service time of charging stations of type k
α	EV penetration level
β	Charging rate
U	Total required number of capacity units to be located, $U = \lceil F/p \rceil$

In this model, we divide the city into a grid with several same cells. A cell is a minimum area that uses to estimate potential charging demand. All the demand cells can be candidate sites for the location. The potential charging demand should be assigned to the nearest available station. All the cells should be covered by one station. As we stated before, we focused on charging stations for traveling EVs that need to get charged in a short time to complete their trips. Therefore, we need to estimate the potential charging demand for traveling EVs in each cell. Lin et al. (2008) developed a location model for hydrogen stations and supposed that any point along a road with higher VMT (vehicle miles traveled) was more likely to be an origin of the refueling trip, which was driven by the notion that '*where you drive more is where you more likely need refueling*'. They quantified the refueling demand of a node with the use of VMT data. Similarly, drivers traveling in any cell may need a charge and they always drive to the nearest available charging station. Based on their assumption, we suppose that a cell with more EVs will produce more charging demand. Because of low EV penetration, we cannot estimate the charging demand through the real data from EVs. However, the overall traffic flow, which is determined by the layout of a city, is an inherent attribute of an urban road network. It is assumed that the prevalence rate of EVs has a little impact on traffic flow distribution and the areas with more conventional vehicles currently will have more EVs in the future.

Based on the above statements and assumptions, the potential charging demand of each cell is estimated as follows. The number of vehicles in each cell (f_i) is derived from the average vehicle traveling speed (v_i) that is obtained by GPS trajectory data from traveling vehicles of the network. Then, the charging demand of a cell i can be estimated by the product of EV penetration level (α), charging rate (β), and the estimated number of vehicles in each cell. The EV penetration level denotes the proportion of EVs in total motor vehicles. The charging rate signifies the probability that a moving EV generates a charging demand. Following the above considerations, the charging demand of cell i is shown in equation (1).

$$d_i = \alpha\beta f_i = \alpha\beta \cdot \rho_i \mu_i g(v_i) \quad (1)$$

Where ρ_i (km/km²) is the density of road network in cell i and μ_i (km²) is the area of cell i . The product of ρ_i and μ_i is the length of roads in cell i . The multi-regime speed-density equation $g(v_i)$ (vehicles/km) is used to estimate flow density of cell i based on average speed v_i . Underwood's exponential model is used for the free-flow regime and Greenberg's logarithmic model is used for the congested regime (Edie, 1961).

The service type (k) provided for a station is either fast plug-in charging or battery swapping. It is assumed that there are several capacity options h for the station in cell i , $h = 1, \dots, H_i$. The capacity unit (p) is the minimum level of the station capacity option, which points to the minimum number of EVs a station can serve per day. If a station chooses capacity option h , it can serve hp EVs per day. The maximum capacity option (H_i) for a station in cell i varies with positions of cells. Considering the space and electric power infrastructure that a station needs, the maximum capacities of cells in the center of a city

would be smaller than cells in suburbs. The total number of capacity units provided by all stations should not exceed the potentially required number U to avoid the resource waste.

3.2 Problem Formulation

The bi-objective charging station location problem, $P(z_1, z_2)$, is formulated as follows. There are two types of decision variables in the model which are x_{jhk} (binary variables, equal to 1 if a charging station of type k is located in cell j with h capacity units) and y_{ijk} (the fraction of vehicles in cell i that is served by charging station of type k in cell j).

The first objective is to minimize the total cost, which is an important issue considered by decision makers, as well as the most common objective of location problems. The cost is usually summarized in literature as a parameter that includes all the necessary cost of building a refueling station. Owing to the need to consider two locating modes and two types of services, the composition of cost needs to be further subdivided. Compared with the gas-station-based location, the random-based location must pay more for construction which includes the building cost and land cost. Moreover, battery swapping equipment is much more expensive than that needed for fast plug-in charging. Therefore, the total cost function includes the construction cost and equipment cost under different combinations of location modes and services. Let c_{jk} denotes the cost of locating a new charging station of type k in cell j with one capacity unit, which is the sum of the construction cost of locating a random-based charging station in cell j and the equipment cost of type k . According to Feng and Evans (2008), the cost of constructing a new station with h capacity units is $h^\delta c_{jk}$. The exponent δ ($0 < \delta < 1$) refers to the rate of cost increases as capacity options rising. The value of δ is less than one because of the economies of scale for constructing a station. The construction cost saved by a gas-station-based location is denoted by e . The 0-1 parameter b_j is used to describe the existing gas station network. The first objective is to minimize total cost as equation (2) shows.

$$z_1 = \sum_{j=1}^N \sum_{k=1}^K \sum_{h=1}^{H_j} (h^\delta c_{jk} - eb_j) x_{jhk} \quad (2)$$

The second objective is to maximize service quality, which directly affects the drivers' willingness to choose the service or to buy EVs, and further affects the sustainable development of the stations. Based on the concept of fuel accessibility presented in Lin et al. (2008), we introduce the notion of electricity accessibility (EA) to measure the service quality of a charging station network. EA is measured by the average time spent by a random driver to complete charging. In this study, EA includes two parts, namely the travel time for a driver to reach a nearest available charging station and the service time spent in the charging station. The travel time is determined by the locations of the charging stations, and the service time is determined by the service types of charging stations. Let t_{ij} represent the travel time from cell i to cell j and s_k represent the service time of charging stations of type k . The second objective is to minimize the average EA as equation (3) shows.

$$z_2 = \sum_{i=1}^N \sum_{j=1}^N \sum_{k=1}^K d_i (t_{ij} + s_k) y_{ijk} / F \quad (3)$$

The model has the six constraints represented by equations (4) - (9). Equation (4) is the coverage constraint that ensures every cell is covered by one station. Equation (5) is the capacity constraint that ensures that the number of EVs served by a station is less than its capacity. When a station of type k with h capacity units has been located in cell j ($x_{jkh}=1$), the number of EVs the station can serve per day should be hp as defined earlier. The amount of assigned charging demand to cell j should not exceed hp . Otherwise, if there is no station located in cell j ($x_{jkh}=0$), the decision variable y_{ijk} should equal to 0. Equation (6) describes the relationship between decision variables that the demand cannot be covered unless a station is installed. Equation (7) ensures that only one service type and one capacity option can be chosen by any station. Equation (8) ensures the total number of capacity units installed cannot exceed the total number required. Equation (9) represents the value ranges of decision variables.

$P(z_1, z_2)$:

$$\text{Min } z_1$$

$$\text{Min } z_2$$

Subject to:

$$\sum_{j=1}^N \sum_{k=1}^K y_{ijk} = 1; \quad i = 1, \dots, N \quad (4)$$

$$\sum_{i=1}^N \sum_{k=1}^K d_i y_{ijk} \leq \sum_{h=1}^{H_j} \sum_{k=1}^K hp x_{jkh}; \quad j = 1, \dots, N \quad (5)$$

$$y_{ijk} \leq \sum_{h=1}^{H_j} x_{jkh}; \quad i, j = 1, \dots, N, \quad k = 1, \dots, K \quad (6)$$

$$\sum_{h=1}^{H_j} \sum_{k=1}^K x_{jkh} \leq 1; \quad j = 1, \dots, N \quad (7)$$

$$\sum_{j=1}^N \sum_{h=1}^{H_j} \sum_{k=1}^K hx_{jkh} \leq U \quad (8)$$

$$x_{jkh} \in \{0,1\}, \quad y_{ijk} \geq 0 \quad (9)$$

Our problem extends the set covering problem, which is known as an NP-hard problem (ReVelle and Eiselt, 2005). In the proposed problem, if we collapse the three dimensions of decision variables into one dimension by avoiding the consideration of capacity options and service types and remove the second objective function, the problem can be converted into a set covering problem. Hence, the proposed problem is more complex than the classic set covering problem. Moreover, the presence of two objectives requires a solution method to find a set of efficient Pareto solutions instead of one optimal solution, which increases the computational cost even more.

4. The hybrid NSGA-II

This section introduces our solution method for the proposed bi-objective location model. As we mentioned in section 2, meta-heuristics techniques are more efficient in solving larger and more sophisticated problems. Our method is based on the well-known NSGA-II. The NSGA-II classifies the population into several fronts of non-dominated chromosomes. The chromosomes belonging to a given front are ranked based on the measure of diversity. The NSGA-II preserves the convergence and diversity of the Pareto set by using non-dominated classification and crowding distance measure, respectively. Typically, the classic NSGA-II is applied to solve discrete problems on which genetic operators can be effectively implemented. In our problem, the discrete decision variables determine the location information of the stations related only to the first objective function. To obtain the Pareto set, the second objective function, which involves continuous decision variables, also needs to be optimized. Hence, we extend the NSGA-II for our model that has both discrete and continuous decision variables.

In the proposed hybrid NSGA-II, we combine the NSGA-II with linear programming (LP) and neighborhood search. The evolution mechanism of NSGA-II is used as the global searching method. The discrete decision variables are decoded as chromosomes so that the genetic operators can be implemented. Two neighborhood search strategies are employed to improve the chromosomes. The continuous decision variables are obtained by constructing and solving a transformed LP problem. The details of the method are provided in the following sections.

4.1 Problem decomposition

The problem is divided into two phases: the location phase and the assignment phase. The location phase is directed at deciding the locations, service types and capacity options of charging stations that are only associated with binary decision variables. A solution of the location phase is encoded as a chromosome and genetic operators are performed on it. The cost objective is calculated based on the chromosome. Given a fixed charging station network, the assignment phase allocates the charging demand of the cells to appropriate charging stations. The phase deals with continuous decision variables. During the assignment phase, the quality objective is fulfilled by solving the transformed LP problem.

1) Location phase: discrete variables encoding and neighborhood search

A solution $[X, Y]$ to the problem includes binary variables X and continuous variables Y obtained from the location and assignment phases respectively. Each solution obtained during the location phase X is coded as three integer strings that represent locations, service types and capacity options (Fig. 1). The length of each string is equal to the number of candidate locations, N , for charging stations. The i -th position of S^1 indicates whether cell i has a charging station (value of 1) or not (value of 0). The i -th position of S^2 indicates that the service type provided by the charging station in cell i is fast plug-in charging (value of 1) or

battery swapping (value of 2). The value of the i -th position of S^3 indicates the capacity option selected by the charging station in cell i , which ranges between 1 and H_i . The coding process starts with a randomly generated S^1 . Next, we randomly choose service types and capacity options in S^2 and S^3 for positions with a value of 1 in S^1 . The chromosome is the combination of the three strings.

S^1 : Location	0	1	1	0	0	1	0	0
S^2 : Service	0	2	1	0	0	1	0	0
S^3 : Capacity	0	1	3	0	0	2	0	0

Fig. 1 Solution decoding

Random generation can guarantee the diversification of solutions, but cannot ensure satisfaction of capacity constraints. Since the total capacity of some solutions may be insufficient or redundant to serve the total charging demand, some improvement methods are needed. Considering the cost objective first, the charging capacity should be concentrated in a few stations in view of the economic scale. As for the quality objective, the charging capacity should be distributed in a dispersed manner to ensure that charging demand can be met at the nearest station in a short time. Because of these competing objectives, this paper utilizes two neighborhood search strategies, namely centralization and decentralization (Feng and Evans, 2008), to convert infeasible solutions to feasible ones.

The centralization strategy involves randomly selecting a position to add/remove all required/redundant capacities. If the required capacity exceeds the maximum capacity of the selected position, the next position is randomly selected until all the required capacity has been realized. If the redundant capacity exceeds the station's maximum capacity, the next station is randomly selected until all the redundant capacity is removed. The decentralization strategy aims to select a position randomly to add/remove one unit of required/redundant capacity. The procedure repeated until all the required/redundant capacity has been added/removed.

2) Assignment phase: LP problem construction

As mentioned above, the assignment phase deals with continuous variables to facilitate the assignment of the charging demand to appropriate charging stations. With fixed X , the optimal solutions Y^* can be obtained by solving a transformed problem $LP(z_2)$, defined as follows:

$LP(z_2)$:

$$\text{Min } z_2$$

$$\text{Subject to: } \sum_{j=1}^N \sum_{k=1}^K y_{ijk} = 1, \quad i \in \{1, \dots, N\} \quad (4)$$

$$\sum_{i=1}^N \sum_{k=1}^K d_i y_{ijk} \leq pS^3(j), \quad j \in \{1, \dots, N\} \quad (10)$$

$$y_{ijk} \leq S^1(j), \quad i, j \in \{1, \dots, N\} \quad k \in \{1, \dots, K\} \quad (11)$$

$$y_{ijk} \geq 0 \quad (12)$$

Where S^1 and S^3 are the first and third strings of chromosome of X . The objective function minimizes EA. Equations (10) and (11) are transformed from equations (5) and (6) when the charging station network has been settled. The $LP(z_2)$ can be quickly and exactly solved by calling the optimization tool. The optimal solution Y^* is the best assignment of the charging demand to the charging stations installed during the location phase. This constructs solution $[X, Y^*]$. The binary decision variables are then decoded as a chromosome while the continuous variables are denoted by the solution of an LP problem.

3) Fitness evaluation

The fitness of a solution is evaluated on the basis of the objective functions. The model has two objectives: total cost z_1 and EA z_2 . As equation (1) shows, the total cost can be calculated by the decoded chromosome. The EA of the solution is the optimal objective value of $LP(z_2)$. Hence, two measures of the fitness of a solution $[X, Y^*]$ are $z_1(X)$ and $z_2(Y^*)$, which evaluate the cost and EA of the solution respectively. The two types of fitness are evaluated by chromosome evaluation and LP problem-solving.

4.2 Evolution mechanism

1) Non-dominated sorting (NDS) and crowding distance sorting (CDS)

Two goals of a multi-objective algorithm are convergence and diversity. The NDS is used to search for convergence, which classifies the population of chromosomes into several fronts. The first front is composed of non-dominated chromosomes in the population. The second is formed by non-dominated chromosomes in the population after removing the chromosomes belonging to the first front. This sorting carries on until every chromosome has been classified into one and only one front. Smaller values of the front number indicate better convergence. The CDS helps maintain the diversity of the population, which sorts the chromosomes in the same front by density. The density is evaluated by the distances between the chromosome and its two adjacent chromosomes. A large value of density indicates greater diversity. These two types of sorting methods rank the chromosomes in the population. (see Deb et al., 2002, for details)

2) Genetic operators

The genetic operators that can be used include selection, crossover, and mutation. We use binary tournament selection to create the mating pool. Once two chromosomes are randomly picked up, the one with the better rank is selected.

Pattern-based crossover is used in the algorithm. Before crossover, a random pattern is generated. Another pattern is automatically generated based on this pattern. Each child is a

combination of the two parts provided by two parents. The crossover process is illustrated in Fig.2.

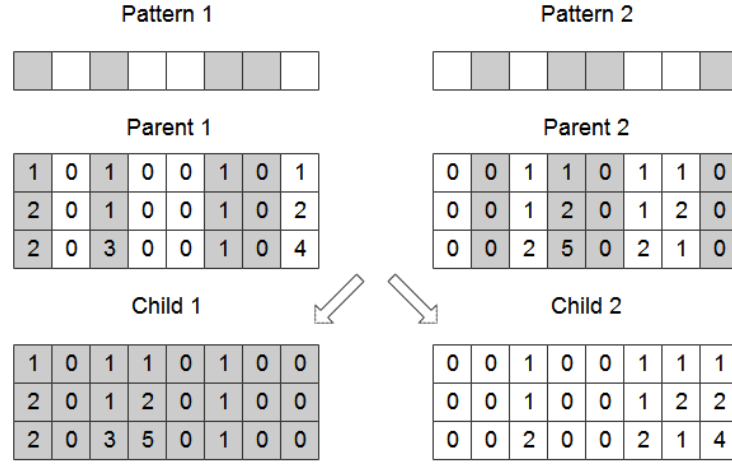


Fig. 2 Crossover

One-point mutation is also utilized in the algorithm. A position of S^1 is randomly selected from a parent. Its value changes from 0 to 1 or from 1 to 0. The corresponding positions of S^2 and S^3 are also changed in a similar way depending on their attributes. The mutation process is illustrated in Fig.3.

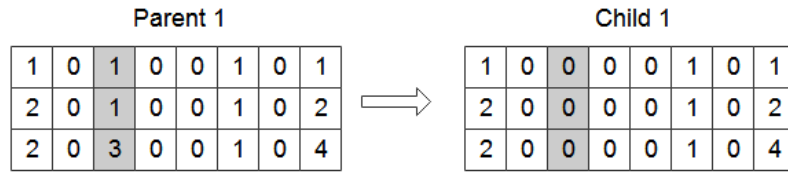


Fig. 3 Mutation

4.3 Procedure of the hybrid NSGA-II

The proposed hybrid NSGA-II combines the NSGA-II with LP and neighborhood search. The general structure of our method is as follows:

Step 1: Divide the problem into a location phase and an assignment phase. Initialize the population by randomly generating chromosomes during location phase.

Step 2: Randomly select neighborhood search strategies to improve infeasible chromosomes in the population and evaluate their fitness z_1 .

Step 3: Solve the set of LP problems and evaluate fitness z_2 .

Step 4: Apply NDS and CDS to rank the chromosomes in the population.

Step 5: Generate offspring using genetic operators including binary tournament selection, pattern-based crossover, and one-point mutation.

Step 6: Repeat steps 2-4 for the offspring chromosomes.

Step 7: Create the next population using the elitist preservation strategy.

Step 8: Repeat steps 5-7 until the stop criterion is met. Select the Pareto solutions in the population to form the output.

The flow chart of the hybrid NSGA-II is given in Fig.4. Improvements over NSGA-II are surrounded by dashed boxes.

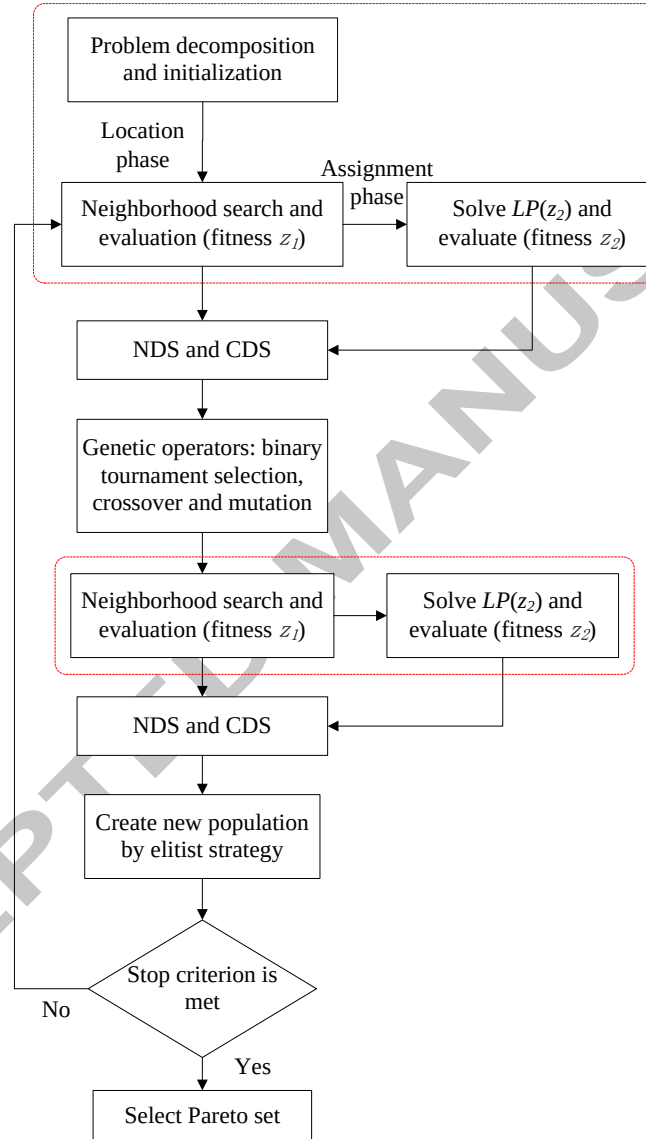


Fig. 4 Flow chart of the hybrid NSGA-II

5. Computational experiments and case study

In this section, we first evaluate the performance of the proposed hybrid NSGA-II algorithm. Then, we present a case study of designing an EV charging stations network for Shenzhen, China based on real GPS trajectory data. The proposed algorithm was coded in the MATLAB software environment (Version R2012a). The LP sub-problems in the hybrid NSGA-II and the single-objective problems in the ϵ -constraint method were solved by

CPLEX 12.6 using default settings. The computational experiments were conducted on a computer with 3.10 GHz speed and 4 GB RAM.

5.1 Performance of the hybrid NSGA-II

As the two objectives (cost and service quality) of the proposed location model are conflicting, there cannot exist a single optimum that simultaneously optimizes both objectives, but a set of Pareto-efficient solutions (Liu, M et al., 2018). The ε -constraint method is one of the most effective approaches for solving bi-objective optimization problems and has been widely used in many real-world applications (Li, Y et al., 2018). It has been proven that exact Pareto front can be obtained by optimally solving all feasible ε -constrained problems (Bérubé, et al., 2009; Abounacer et al., 2014). Further, the ideal and nadir points can be obtained during this method, which can define the lower and upper bounds on the values of the two objectives. Therefore, we compare our proposed hybrid NSGA-II algorithm with the classic ε -constraint method to test the performance of our algorithm.

1) ε -Constraint method

The ε -constraint method (Vira and Haimes, 1983) is one of the scalarization techniques for solving the bi-objective optimization problem. The basic idea of the ε -constrained method is to transform the original bi-objective model into a set of ε -constrained problems. The ε -constrained problems are constructed by selecting one of the objectives as the only objective to be optimized while the other is treated as additional inequality constraint bounded by a set of ε -vectors. For the bi-objective problems in this paper, the ε -constrained problems are defined by $P(z_1, \varepsilon_2)$ and $P(z_2, \varepsilon_1)$ as follows:

$$\begin{aligned} & \min z_1 \\ & z_2 \leq \varepsilon_2 \\ & \text{s.t. Equations (4)-(9)} \\ & \min z_2 \\ & z_1 \leq \varepsilon_1 \\ & \text{s.t. Equations (4)-(9)} \end{aligned}$$

The ε -constraint method starts by finding the ideal point (z_1^I, z_2^I) and nadir point (z_1^N, z_2^N) , which are the lower and upper bounds on the objective values of efficient solutions (proved by Bérubé et al., 2009). The two points are calculated as follows, where $s=[X, Y]$ is a solution and Φ is the feasible region.

Ideal point:

$$z_1^I = \min\{z_1(s) \mid s \in \Phi\}; \quad z_2^I = \min\{z_2(s) \mid s \in \Phi\}.$$

Nadir point:

$$z_1^N = \min\{z_1(s) \mid z_2(s) = z_2^I, s \in \Phi\}; \quad z_2^N = \min\{z_2(s) \mid z_1(s) = z_1^I, s \in \Phi\}.$$

Extreme solutions:

(z_1^I, z_2^N) and (z_2^I, z_1^N)

The ideal point and nadir point are illustrated in Fig. 5. The Pareto front is included between the two extreme points: (z_1^I, z_2^N) and (z_1^N, z_2^I) . All the other Pareto-efficient solutions would fall in the rectangular region formed by the two extreme solutions (Bérubé et al., 2009). In order to obtain the exact Pareto front, we need to know all possible values of ε_j , which is actually an interval determined by the ideal point and nadir point. Ideally, we should generate the exact Pareto set which requires solving a large number of ε -constrained problems. As the continuous nature of ε_j , it is impractical and unnecessary for decision makers (Li, Y et al., 2018).

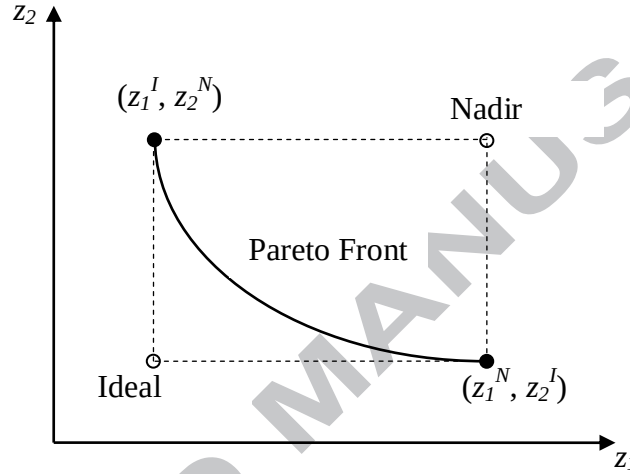


Fig. 5 Illustration of the Ideal and Nadir point (Bouziaren and Aghezzaf, 2018)

Therefore, in this paper, we constructed a limited number of ε -constrained problems $P(z_1, \varepsilon_2)$ and $P(z_2, \varepsilon_1)$ with $z_1^I \leq \varepsilon_1 \leq z_1^N$ and $z_2^I \leq \varepsilon_2 \leq z_2^N$, where ε_j is increased by a constant value throughout the algorithm. The ε -constrained problems are optimally solved by CPLEX 12.6. Finally, a set of non-dominated solutions is selected from the union of the two candidate solution sets obtained by solving the two ε -constrained problem sets. By this procedure, we can obtain some good representatives of the total exact Pareto set, which are used to compare with solutions obtained by the proposed method.

2) Computational results

Computational experiments are conducted on some randomly generated instances to evaluate the proposed solution methods. Instances are generated based on three model parameters: the scale of the network, construction cost distribution and charging demand distribution. The scale of the network is represented by the number of cells in the grid approximating the urban network (N). We generated instances of 9 different sizes of network scale ranging from $N=9$ to $N=300$. Two types of construction cost distribution are considered while generating instances (type C1 and C2). In type C1, the costs in all cells have uniformly distributed values drawn from a fixed range. In type C2, the central cells of the network have a higher cost than outer cells. The charging demand is estimated based on the traffic situation of the network (see Section 3). In a similar way, two types of transportation situations, T_1 and

T_2 , are considered. In type T_1 , the speeds in all cells have uniformly distributed values drawn from a fixed range. In type T_2 , the central cells of the network have higher speeds than those at the outer cells. Considering all scale possibilities, the construction cost distributions, and the transportation situations, there are 36 problem instances. Each instance is named by NC#T#, for example, 8C1T1 represents the instance that has eight cells in the network, type C1 of construction cost distribution and type T1 transportation situation.

The only parameter used in the ε -constraint method is the number of intervals γ which determines the number of iterations or constrained problems solved in the algorithm. After tuning, we let γ equal to 25 that means 50 constrained problems solving for each instance. As for the hybrid NSGA-II algorithm, the following parameters need to be adjusted: the probabilities for crossover (P_c) and mutation (P_m), the number of population, num_POP , and the number of generations, num_Gen). After tuning, we set $P_c=0.9$, $P_m=0.1$, $num_POP=50$ and $num_Gen=200$.

To illustrate the convergence of the proposed hybrid NSGA-II algorithm, we compare the best solutions obtained by the proposed method with the ideal points that are the lower bounds for objective values. The ideal point for each small instance is obtained by solving two single-objective problems using CPLEX, which are listed in columns 2 and 5 of Table 3. The best values of the two objectives obtained by the proposed NSGA-II are listed in columns 3 and 6. The average gaps between the lower bounds and the best values obtained by the proposed NSGA-II for two objectives are 0.04% and 0.67% respectively, which indicate a good performance of the proposed method in terms of convergence.

Table 3 Comparison between the ideal points and the solutions of hybrid NSGA-II

Instance	Objective1:Cost			Objective2: EA		
	Ideal	NSGA-II	Gap	Ideal	NSGA-II	Gap
9C1T1	72.797	72.797	0.00%	0.252	0.252	0.00%
9C1T2	76.114	76.114	0.00%	0.238	0.238	0.00%
9C2T1	65.968	65.968	0.00%	0.260	0.260	0.00%
9C2T2	65.090	65.090	0.00%	0.242	0.242	0.00%
16C1T1	140.524	140.524	0.00%	0.214	0.214	0.00%
16C1T2	107.034	107.034	0.00%	0.220	0.220	0.00%
16C2T1	93.442	93.442	0.00%	0.232	0.232	0.00%
16C2T2	93.861	93.861	0.00%	0.225	0.225	0.00%
25C1T1	166.732	166.732	0.00%	0.213	0.215	0.87%
25C1T2	136.587	136.587	0.00%	0.213	0.214	0.69%
25C2T1	152.934	152.934	0.00%	0.205	0.206	0.10%
25C2T2	121.609	121.609	0.00%	0.216	0.218	0.91%
36C1T1	243.599	243.599	0.00%	0.200	0.206	3.15%
36C1T2	203.601	203.601	0.00%	0.202	0.202	0.00%
36C2T1	258.380	259.911	0.59%	0.196	0.201	2.76%
36C2T2	184.139	184.139	0.00%	0.204	0.208	2.17%
Average			0.04%			0.67%

To further evaluate the performance of the proposed method in terms of convergence, diversity and computational time, we use four widely used metrics that are shown as follows.

- The convergence metric Ω is the size of space covered (Zitzler and Thiele, 1998). The standardized size of the space covered (Villegas et al., 2006) is used in this paper. Its value ranges from 0 to 1. A bigger value of Ω indicates better convergence.
- The diversity metric Δ we used was originally introduced by (Deb et al., 2002). It measures the distribution and extension of the Pareto set. For other sets, the value of the metric would be greater than 0; smaller is better.
- The number of solutions found is the third metrics, which counts the number of solutions in the obtained Pareto set, $|PS|$.
- Computational time is the fourth metric.

Table 4 provides the comparison results between the ε -constraint method and the NSGA-II on small-scale instances. We compare the two methods using the four multi-objective optimization metrics. As Table 4 shows, the two methods perform almost equally in convergence and the ε -constraint method has better performance on diversity for most small-scale instances. Therefore, the quality of the Pareto solutions obtained by the ε -constraint method is a little better than that by the hybrid NSGA-II. However, for other metrics, the hybrid NSGA-II has some advantages. The hybrid NSGA-II can always obtain more Pareto solutions than the ε -constraint method. In terms of computational time, the ε -constraint method shows an exponential increase while the hybrid NSGA-II is very stable. Every ε -constrained problem uses CPLEX to find its optimal solution, which makes the ε -constraint method time-consuming.

Table 4 Comparison between the ε -constraint method and the hybrid NSGA-II on small-scale instances

Instance	ε -constraint method				Hybrid NSGA-II			
	Ω	Δ	$ PS $	Time(s)	Ω	Δ	$ PS $	Time(s)
9C1T1	0.5851	0.6738	10	3	0.5856	0.9445	14	81
16C1T1	0.6499	0.6969	32	30	0.6509	0.7133	40	89
25C1T1	0.6173	0.5749	42	288	0.6185	0.6134	42	91
36C1T1	0.7253	0.4842	43	2219	0.7247	0.6152	46	97
9C1T2	0.6114	0.4462	8	3	0.6119	0.9136	14	75
16C1T2	0.6745	0.9731	21	12	0.6752	1.1015	33	91
25C1T2	0.7083	0.7188	31	61	0.7097	0.8499	41	92
36C1T2	0.6978	0.6968	40	1052	0.6971	0.6985	45	96
9C2T1	0.5796	0.4811	8	2	0.5796	0.4811	8	81
16C2T1	0.6489	0.6246	17	9	0.6501	0.7059	17	91
25C2T1	0.7031	0.7421	46	298	0.7023	0.6822	46	91
36C2T1	0.7150	0.5402	45	2098	0.7245	0.5743	45	92
9C2T2	0.6804	0.6642	11	3	0.6810	0.8111	11	81
16C2T2	0.7281	0.7576	14	10	0.7288	1.0275	21	89
25C2T2	0.7394	0.6816	33	34	0.7407	0.8193	37	92
36C2T2	0.7419	0.5568	37	1129	0.7568	0.7029	40	99

Fig.6 shows the efficient Pareto fronts of four small instances obtained by the two solution methods. Here the cost is presented as a percentage of minimal cost that is calculated when all the charging demand is covered without any requirement for electricity accessibility. It can be observed that Pareto fronts obtained by the proposed method are as good as this classic method.

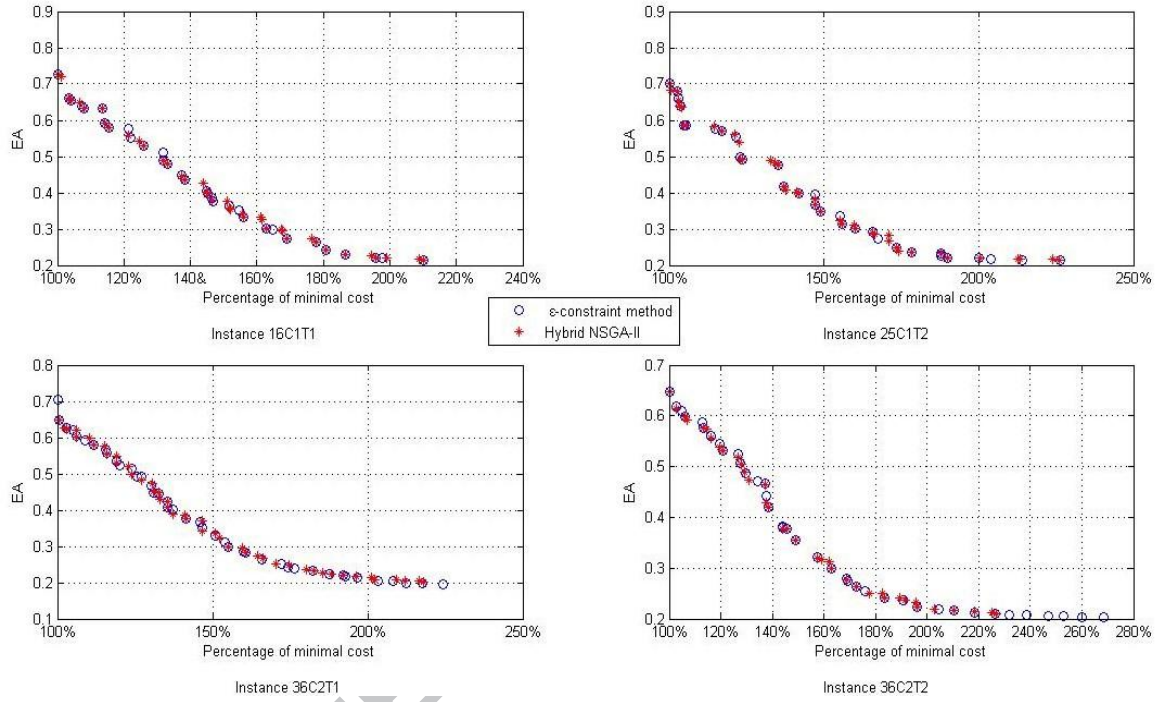


Fig. 6 Efficient Pareto fronts of four instances

Table 5 Results of the hybrid NSGA-II on large-scale instances

Instance	Hybrid NSGA-II				Time
	Ω	Δ	$ PS $		
100C1T1	0.7349	0.6626	47		142
150C1T1	0.7441	0.6017	48		217
200C1T1	0.7492	0.7001	48		272
250C1T1	0.7545	0.7033	49		348
300C1T1	0.7609	0.7175	50		454
100C1T2	0.7462	0.6310	47		138
150C1T2	0.7389	0.6615	48		207
200C1T2	0.7590	0.6729	48		278
250C1T2	0.7746	0.7286	48		352
300C1T2	0.7600	0.7002	49		485
100C2T1	0.7758	0.6667	45		141
150C2T1	0.7700	0.6604	48		216
200C2T1	0.7770	0.6913	48		276
250C2T1	0.7749	0.6969	50		375
300C2T1	0.7641	0.7477	49		482
100C2T2	0.7823	0.6847	47		138

150C2T2	0.7931	0.7146	47	210
200C2T2	0.7832	0.6745	49	289
250C2T2	0.7730	0.7689	48	375
300C2T2	0.7639	0.7095	49	496

For large-scale instances, only the results of the hybrid NSGA-II are listed in Table 5 because these instances cannot be solved by the ε -constraint method within a reasonable amount of time. The results indicate that the hybrid NSGA-II is time-saving and performs stably in large-scale instances.

Overall, both the ε -constraint method and the hybrid NSGA-II obtain efficient Pareto sets and fronts. The ε -constraint method has a slight advantage in terms of the quality of obtained Pareto sets. Since the hybrid NSGA-II can obtain more Pareto solutions in a much shorter time while addressing large-scale instance, the latter is more appropriate in practice. Therefore, we use the hybrid NSGA-II in the case study.

5.2 Application to the case in Shenzhen

Shenzhen is one of the leading cities in terms of possessing and promoting EVs in China. At the time of our study, there are a few thousand EVs in Shenzhen including electric buses, taxis, and private cars. Electric charging stations are relatively few now, which is the major barrier for drivers choosing EVs. This case involves designing an EV charging station network in Shenzhen for current and future potential use. The city is approximated by a grid with the estimated charging demand noted for each cell. A set of possible solutions is obtained by the proposed method. Two objectives: cost and EA, are considered during decision-making. The impacts of some parameters are discussed in the sensitivity analysis part.

1) Data and processing methods

The proposed model was used to design an EV charging stations network for Shenzhen, China. The data used in the case study consisted of the one-month GPS trajectory data collected from vehicles in Shenzhen. Our data was from the database of Yiliu Company, which offers vehicle fleet management and transportation information service to China's freight industry. It has more than 500,000 freight vehicles registered in the system. When vehicles are traveling, the database collects information of vehicles every few seconds. Each record comprises vehicle data including vehicle ID, date and time, longitudinal and latitudinal coordinates, speed, and direction, (see Table 6). According to the database, the number of active vehicles in Shenzhen is larger than 10,000 per day and the average number of records is 4320 per vehicle per day. This huge volume of data enables a good understanding of the urban traffic situation.

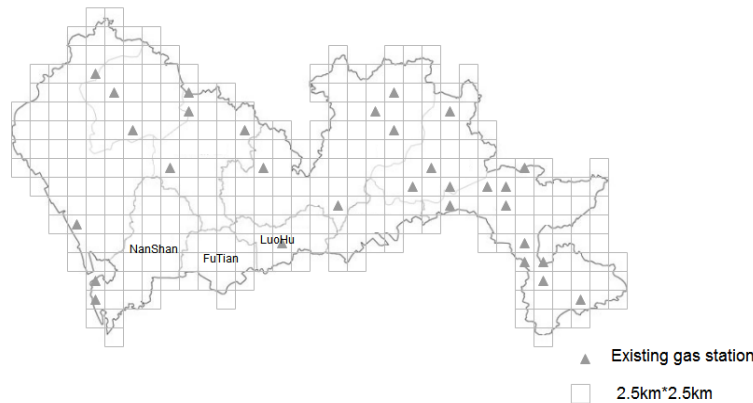
Table 6 One record of GPS trajectory data

Vehicle ID	Date and time	longitude	Latitude	mileage(km)	velocity(km/h)	direction
376	10/25/2015 11:27:40	114.5888	22.7752	70825.46	57	291

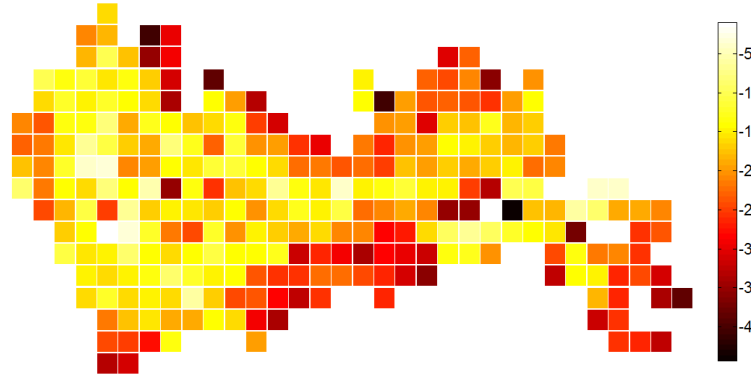
Although GPS trajectory data can reflect the transportation situation in a city, it cannot be used directly by the proposed model. A method is needed to extract useful information from the raw data. In the event, a three-stage method was developed to estimate the charging demand from GPS trajectory data. The details are as followings:

- Preprocessing. The major work in this stage is to extract valid data, which involves removing records that are out of the range or invalid.
- Gridding. The study area is divided into several small cells with the same length and width. The record points in cells are clustered.
- Quantification. The average speed of each cell (v_i) is estimated using hundreds of record points. The estimated charging demand in each cell (d_i) is computed by equation (1).

For this case study, the city is approximated by a grid with 325 cells (2.5km per unit length), among which there are $N=287$ valid demand cells and candidate sites. A lot of gas stations owned by different companies exist in the city. The locations of existing gas stations are decided by the company that decision makers plan to incorporate with. In this case, we randomly choose 30 cells that have existing gas stations. The grid network of Shenzhen is shown in Fig.7a. The three named areas are downtowns of the city. After processing the data, the number of EVs in each cell is estimated as shown in Fig.7b (EV penetration level = 0.1). Multiplying by the charging rate determines the charging demand in each cell. The travel time between the two cells is calculated using the average travel speed during the process. Thus, the inputs of the model are obtained. These include the number and the geographical coordinates of demand cells and candidate sites, the existing gas station network, charging demand, and travel time matrix of the network.



a) Grid network of Shenzhen



b) The distribution of estimated numbers of EVs in Shenzhen ($\alpha = 0.1$)

Fig. 7 Illustration of the case and data

2) Results

Given the input data, an EV charging stations network was designed using the proposed model and method. The model has several parameters. The capacity unit of the charging station was set at 60, which signified the minimum number of vehicles a station serves per day. The EV penetration level was set at 0.1. The charging rate was set at 0.5. The service times for fast plug-in charging and battery swapping were set at 0.5 hours and 0.15 hour, respectively. Since the parameters associated with cost were fluctuating, we just used relative values and presented the percentage results. The parameters of the hybrid NSGA-II are the same as the previous computational experiments.

The Pareto set obtained with 50 non-dominated solutions is shown in Fig.8. The results illustrate the tradeoff between the cost and EA. With the minimum input cost, the designed charging station network offers the lowest quality with EA of 0.94 hours. The best EA obtained is 0.42 hour, while the cost rises to 268.47% to achieve the highest quality. Between these two extreme situations, there are many choices for decision makers to balance the cost and quality.

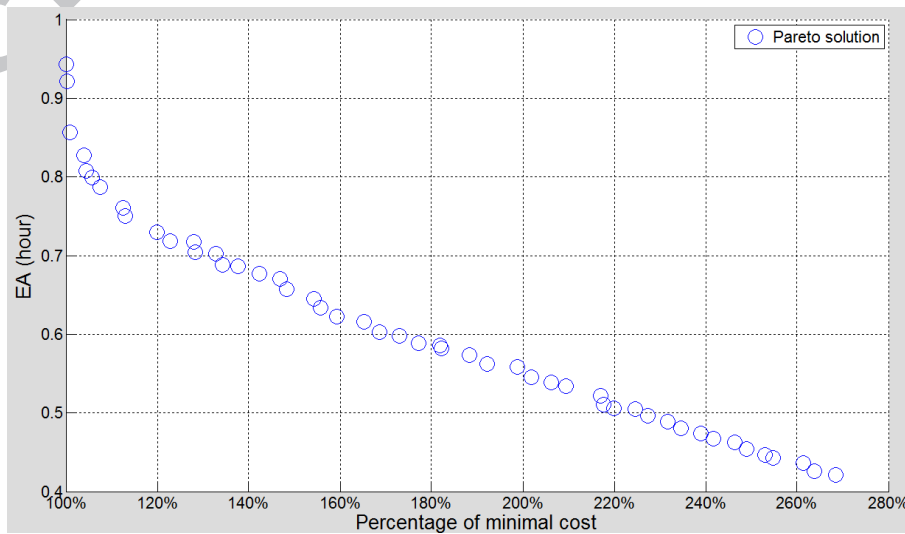


Fig.8 Pareto set of the case

Table 7 Several solutions in the obtained Pareto set

Solution	Cost	EA(hour)	Average capacity	Num_station	Num_fast	Num_battery
Sol1	100.00%	0.94	3.8	12	12	0
Sol5	104.52%	0.81	2.6	18	18	0
Sol10	119.94%	0.73	1.8	25	24	1
Sol15	134.27%	0.69	1.8	25	22	3
Sol20	154.22%	0.64	1.7	27	21	6
Sol25	173.02%	0.60	1.8	25	18	7
Sol30	192.23%	0.56	2.0	23	13	10
Sol35	217.01%	0.52	1.5	30	15	15
Sol40	231.71%	0.49	2.0	23	7	16
Sol45	249.03%	0.45	2.1	22	4	18
Sol50	268.47%	0.42	1.8	26	3	23

Several solutions in the obtained Pareto set are numbered from minimum cost to maximum cost and several of them are listed in Table 7. Two objective values are listed in the *Cost* and *EA* columns, which verify the tradeoff between cost and quality. Sol1 is the solution with minimum cost and sol50 is the solution with minimum EA. *Average capacity* is the mean of the capacity options of all the charging stations in the solution. Its value decreases from Sol1 to Sol50 while displaying slight fluctuations between the two extreme solutions. A Higher value of average capacity represents a centralization of the resource, which brings poorer service quality and lower cost because of economies of scale while constructing the station. Conversely, a smaller value of average capacity points to a good distribution of resource, i.e., it offers better service and higher cost. Consequently, the total number of stations has a generally rising trend, as shown in the next column *Num_station*. The number of fast-charging stations (*Num_fast*) and the number of battery-swapping stations (*Num_battery*) summarize the character of stations' service type. Fast plug-in charging has lower cost and longer service time, which is contrary to the state in battery swapping. Hence, solutions with less cost are likely to involve fast plug-in charging while those with less EA prefer battery swapping. The trends are depicted in Fig.9.

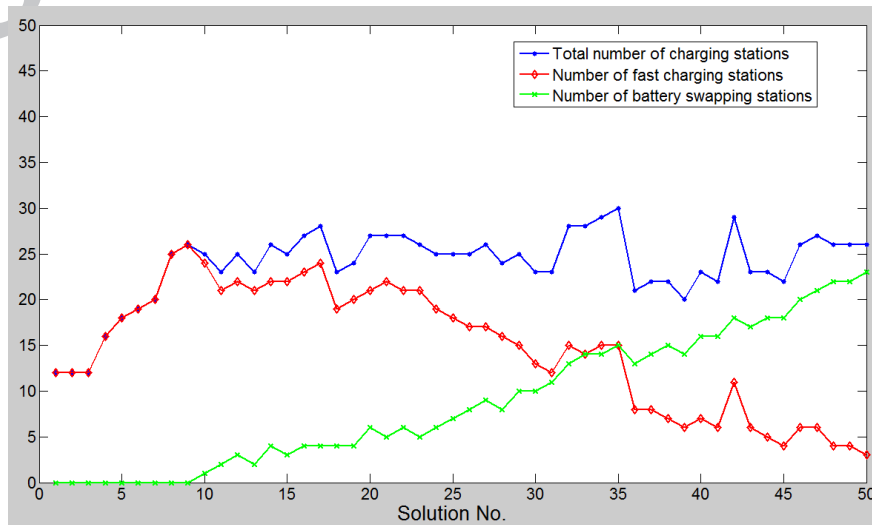
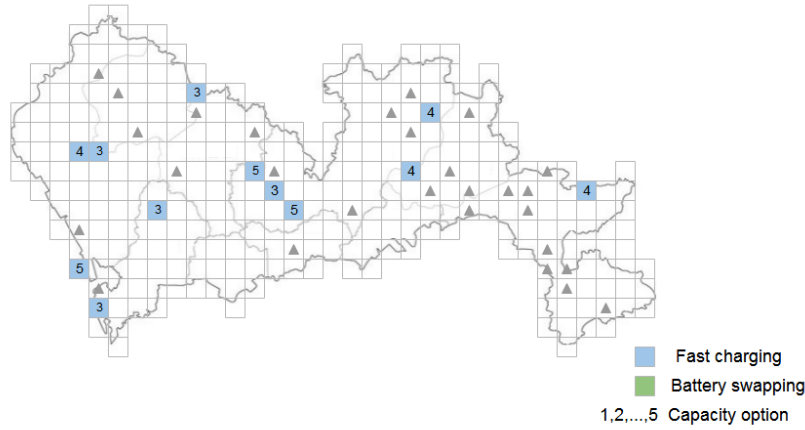
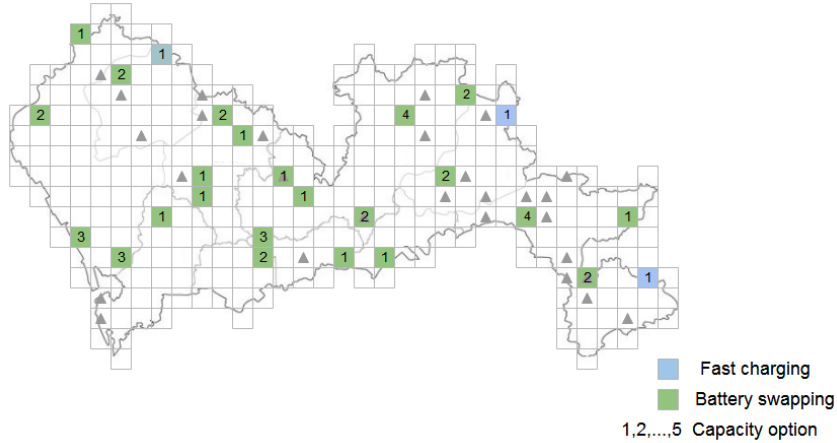


Fig.9 Variation of service types of stations

Fig.10 intuitively illustrates the two extreme solutions on the grid map. Cells with colors are the locations of charging stations. Different colors represent different service types. Numbers in the cells denote capacity options of stations. Referring to Fig.10a, all the stations provide fast charging. To save money, all stations choose the higher capacity options ($h \geq 3$). Thus, it takes longer for drivers to find stations and complete charging. On the contrary, in Fig.10b, to achieve high service quality, most of the stations are battery-swapping stations who choose smaller capacity options. The number of charging stations increases from 12 to 26 with increasing decentralized capacity. This suggests that the higher quality is more expensive.



a) Location with minimum cost & lowest quality



b) Location with maximum cost & highest quality

Fig.10 Grid maps of possible charging station locations

In summary, a set of possible solutions is presented to the decision-maker. The cost and the quality level achieved (EA) are provided to him/her for reference. A decision maker preferring cost may choose the solution with minimum cost. Conversely, if a decision maker is in the government who is more concerned with the service quality to promote EV penetration level, the solution with the minimum EA would be selected. Besides the two extreme solutions, several other solutions can be used by different decision-makers. Once the

decision-maker has chosen a solution, we can offer complete information to build an EV charging station network including the locations, service types and capacity options.

3) Sensitivity analysis

In this section, we discuss how parameters affecting the objective values were set. Sensitivity experiments were conducted on the following variables.

- EV penetration level (α). EV penetration level directly decides the charging demand of the city which is the key input data of the model.
- Service time (S_k). The service time of fast plug-in charging and battery swapping would change with the development of power and charging technologies. The service quality is highly dependent on this parameter.
- Capacity unit (p). Its value is set at the minimum number of vehicles a station can serve per day, which measures the basic scale of charging stations.

Three scenarios were constructed and their Pareto sets compared with the original scenario discussed in section 5.2.1. The three scenarios were 1) doubling the EV penetration level to 0.2, 2) reducing the service time of two service types by 20% and 3) doubling the capacity unit to 120 vehicles per day. Figs. 11-13 show the comparisons between the original scenario and the three scenarios respectively.

Fig.11 shows the comparison results between scenario 1 and the original scenario, which depicts the effect of the EV penetration level on the Pareto sets. Doubling the value of EV penetration means doubling the charging demand. As the figure illustrates, the cost grows up. This indicates that the government and the investors need to continue investing in new charging stations to satisfy increasing charging demand. In the new scenario, the average *Num_station* of all solutions in the new Pareto set is 43.2, which is larger than 23.7 in the original Pareto set. Conversely, the average value of EA in the new Pareto set is 0.56 hour, which is smaller than 0.61 hour in original Pareto set. The results can be explained by economies of scale in the charging station network, which attracts more drivers to replace their petroleum-based vehicles by EVs.

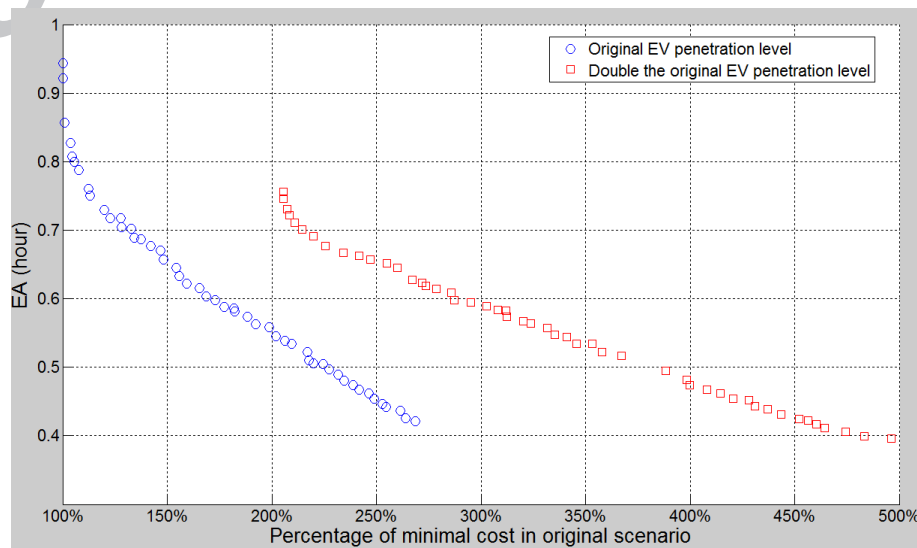


Fig.11 The effect of the EV penetration level on the Pareto sets

Fig.12 depicts the effect of a change in service time on the Pareto sets. The values of the worst and best EA have improved 18.60% and 4.75% respectively, but the costs needed have decreased by 2.49% and 1.23%. With the same EA, 0.7 hours, for example, the cost saved through service time improvement is 19.88%. It is therefore concluded that the advance in power technology has positive effects on both cost and quality.

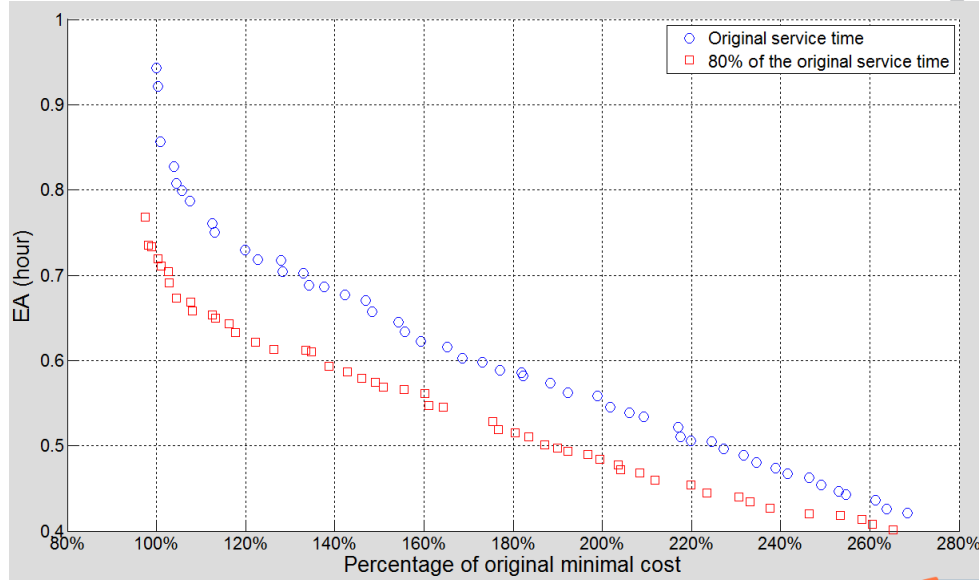


Fig.12 The effect of the service time on the Pareto sets

Fig.13 illustrates the effect of the capacity unit. The Pareto sets of scenario 3 and the original scenario are compared. Doubling the capacity unit increases the basic scale of charging stations, which cuts down the number of charging stations required. Greater economies of scales on constructing a station reduce the cost needed to cover all the charging demand. However, it takes longer for a random driver to find a charging station. The average value of EA in the new Pareto set increases to 0.68 hours (from the 0.61 hours in the original Pareto set). Hence, the increase of the basic scale of charging stations can save money but worsens the service quality.

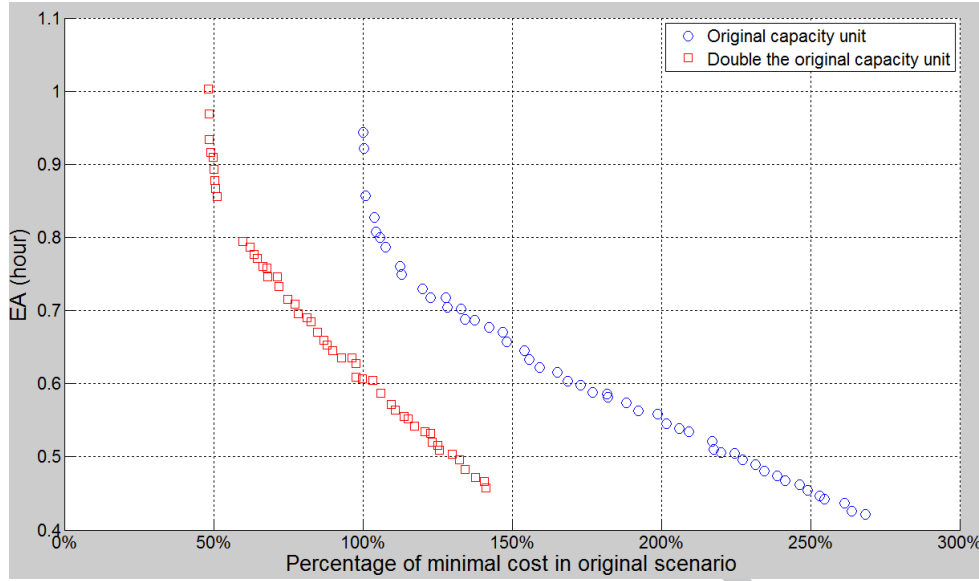


Fig.13 The effect of the capacity unit on the Pareto sets

5 Conclusions

In this paper, the location problem of EV charging stations has been studied. The purpose has been to design an EV charging station network in the city to encourage more drivers to choose EVs. We proposed a cell-based location model that used GPS trajectory data drawn from the thousands of vehicles on roads to estimate future charging demand. Two types of services (fast plug-in charging and battery swapping) and locating modes (gas-station-based and random-based) have been considered. The problem is formulated as a bi-objective MILP optimization model with one objective related to total cost and the other related to service quality. The results of the model provide decision-makers with a set of possible solutions under different preferences.

Secondly, the hybrid NSGA-II has been proposed to determine an efficient Pareto set for the problem, which combines the NSGA-II with LP and neighborhood search. The proposed method extends the NSGA-II for the problem that has both discrete and continuous decision variables. The evolution mechanism of NSGA-II is used as the global searching method. The discrete decision variables are decoded as chromosomes that the genetic operators are implemented. Two neighborhood search strategies are employed to improve the random chromosomes. The continuous decision variables involved are obtained by constructing and solving a new LP problem. We compare the hybrid NSGA-II against the traditional ϵ -constraint method. The results show that the hybrid NSGA-II can obtain more Pareto solutions in a much shorter time for large-scale instances with good quality.

Finally, a case study involving designing an EV charging stations network for Shenzhen, China, based on GPS trajectory data from thousands of vehicles has been presented. The tradeoff between cost and quality is demonstrated and an efficient set of possible solutions is obtained. The impacts of certain parameters have been illustrated in the sensitivity analysis. It has been found that the optimization of the cost objective leads to a smaller number of

charging stations with a centralized capacity and a higher proportion of fast-charging stations. The optimization of the quality objective results in a larger number of charging stations with decentralized capacity and a higher proportion of battery-swapping stations.

Future work includes extending our model by considering more criteria such as fairness of the charging station network and utilization of stations. New objective functions and decision variables created to this end will need to be added to the mathematical model. This would increase the difficulty of obtaining the Pareto set. More efficient solution methods are therefore needed. In addition, the average speed used in this paper is insufficient to generate charging demand since traffic volume in a city has a significant variance throughout the day. Future research can adopt a dynamic perspective while designing the location model.

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Highlights

1. This paper addresses the problem of designing a charging station network in the city with low EV penetration rate.
2. We present a cell-based model to decide locations, capacity options, and service types for EV charging stations.
3. The hybrid evolutionary algorithm is developed and evaluated.
4. A case study of Shenzhen, China is presented with GPS trajectory data from thousands of vehicles.